Base

Comparison

We compute the number of Hispanic White alone people in each tract in Georgia using data from the 5 year ACS (2022 vintage) and the 2020 Decennial Census.

For each, we first get the number of White alone people using the following variables:

- **ACS**: B01001A_001E ("Estimate!!Total:" from "Sex by Age (White Alone)")
- **Decennial Census**: P1_003N ("!!Total:!!Population of one race:!!White alone")

We then get the number of non-Hispanic White alone people using the following variables:

- **ACS**: B01001H_001E ("Estimate!!Total:" from "Sex by Age (White Alone, Not Hispanic or Latino)")
- **Decennial Census**: P2_005N ("!!Total:!!Not Hispanic or Latino:!!Population of one race:!!White alone")

Compute the estimates:

In[122]:=
```
pathACS = {"data", "2022", "acs", "acs5"};
```

In[36]:=
```
pathACS = {"data", "2022", "acs", "acs5"};
pathDec = {"data", "2020", "dec", "pl"};
```

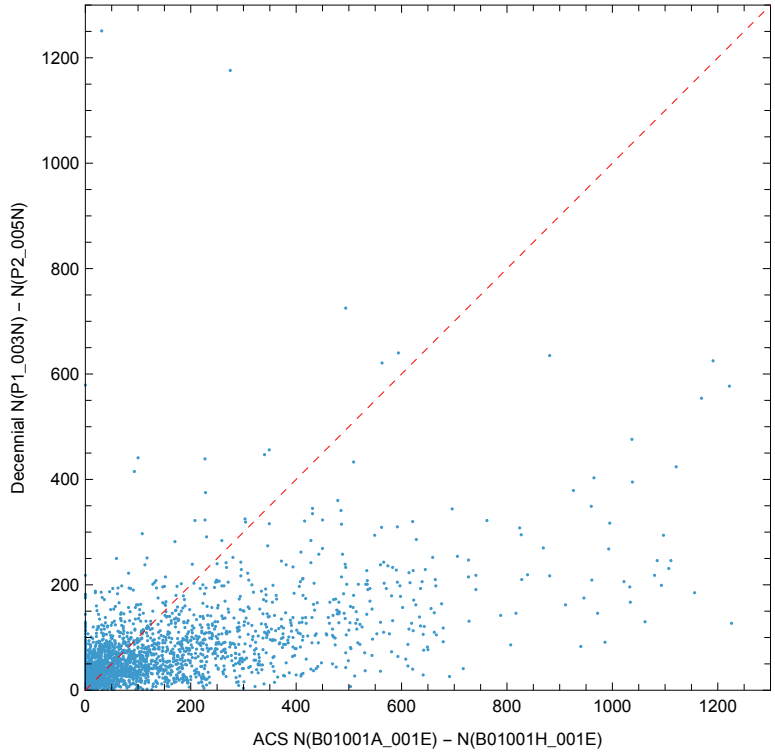
In[35]:=
```
fipsCode = Georgia, United States ADMINISTRATIVE DIVISION [ FIPS code ]
```

Out[35]=
```
13
```

In[39]:=
```
tractCountsACS =
  tractValues[{"B01001A_001E"}, fipsCode, pathACS] -
   tractValues[{"B01001H_001E"}, fipsCode, pathACS];
tractCountsDec =
  tractValues[{"P1_003N"}, fipsCode, pathDec] -
   tractValues[{"P2_005N"}, fipsCode, pathDec];
```

The results are very different from the ACS and Decennial Census:

Out[105]=

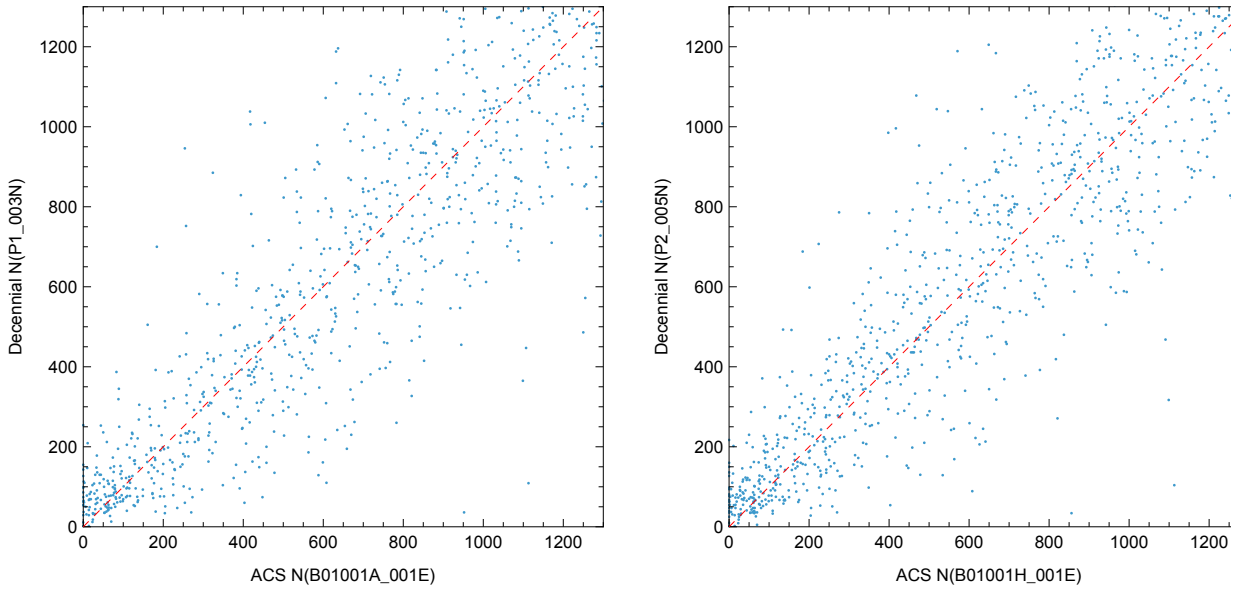


Their distributions are different too. The ACS has far more zero inflation, but also has a fatter tail:

In[42]:=
```
Histogram[{
  tractCountsACS,
  tractCountsDec
 },
 {10},
 ChartLegends → {"ACS", "Decennial"},
 Frame → True, FrameLabel → {"non-hispanic white alone population", "# of tracts"}
]
```

Comparing the direct variables from the ACS and Decennial Census on their own, they don't look too dissimilar:

Out[49]=



Statewide

The state-level numbers also disagree:

In[67]:=
```
stateValue[vars_, fipsCode_, path_] :=
 Total@ToExpression@Normal@censusQuery[path, {"get" → StringRiffle[vars, ","],
      "for" → "state:" <> fipsCode}][[1, vars]]
```

ACS value:

In[64]:=
```
stateValue[{"B01001A_001E"}, fipsCode, pathACS] -
 stateValue[{"B01001H_001E"}, fipsCode, pathACS]
```

Out[64]=

374 864

Decennial value:

In[69]:=
```
stateValue[{"P1_003N"}, fipsCode, pathDec] -
 stateValue[{"P2_005N"}, fipsCode, pathDec]
```

Out[69]=

193 327

Links

https://data.census.gov/table?q=ACSDT5Y2022.B01001A&g=040XX00US13
https://data.census.gov/table?q=ACSDT5Y2022.B01001H&g=040XX00US13

https://data.census.gov/table/DECENNIALPL2020.P1?q=P1&g=040XX00US13
https://data.census.gov/table/DECENNIALPL2020.P2?q=P2&g=040XX00US13

State-level

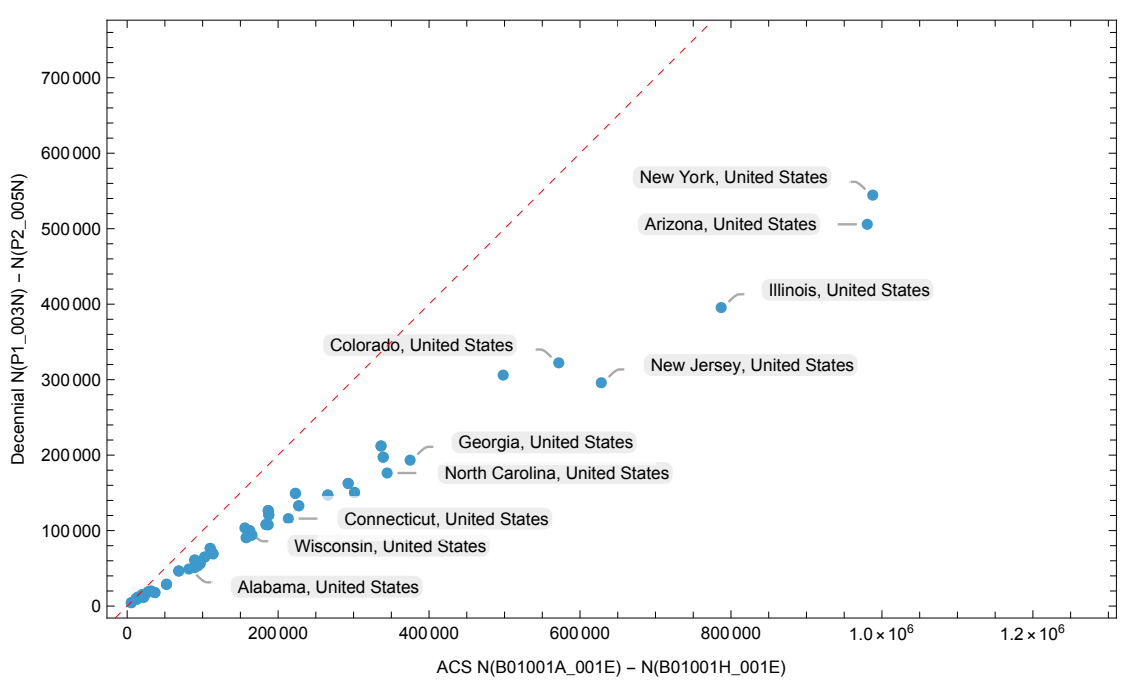
In[ ]:=
```
pathACS = {"data", "2022", "acs", "acs5"};
```

In[124]:=
```
stateCountsACS =
  allStateValues[{"B01001A_001E"}, pathACS] -
   allStateValues[{"B01001H_001E"}, pathACS];
stateCountsDec =
  allStateValues[{"P1_003N"}, pathDec] -
   allStateValues[{"P2_005N"}, pathDec];
```

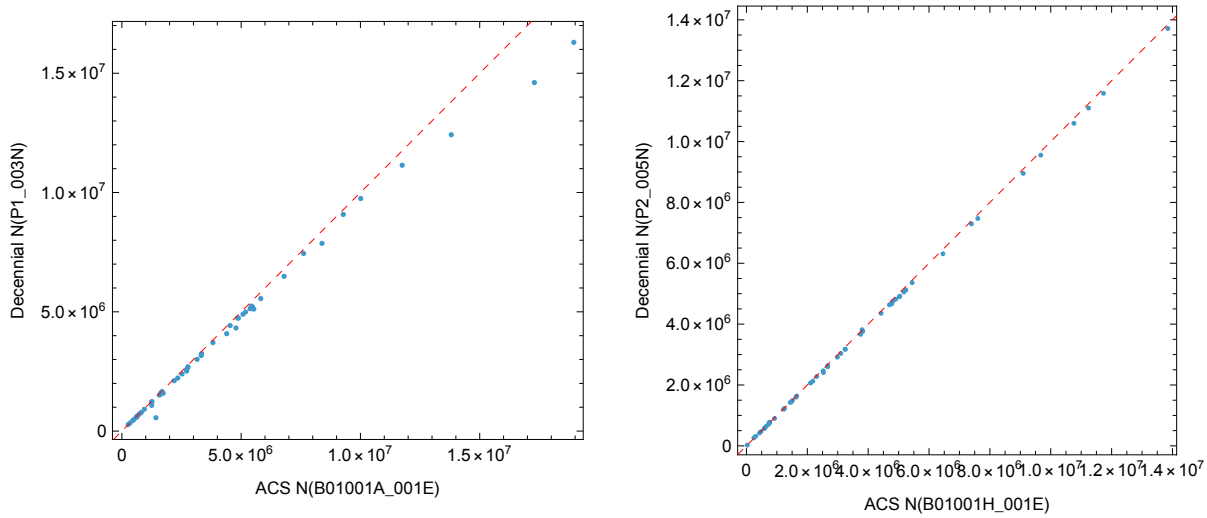
In[120]:=
```
fipsStates = AssociationMap[Reverse, EntityValue[
    all US states with District of Columbia ADMINISTRATIVE DIVISIONS, "FIPSCode", "EntityAssociation"]];
```

In[126]:=
```
ListPlot[
 KeyMap[fipsStates, Merge[{stateCountsACS, stateCountsDec}, Identity]],
 Epilog → {Red, Dashed, InfiniteLine[{0, 0}, {1, 1}]},
 AspectRatio → Automatic,
 (*PlotRange→{{0,1300},{0,1300}},*)
 Frame → True,
 FrameLabel →
  {"ACS N(B01001A_001E) - N(B01001H_001E)", "Decennial N(P1_003N) - N(P2_005N)"}
]
```

Out[126]=



Out[116]=



Different vintages

2022 1-year

In[127]:=

```
pathACS = {"data", "2022", "acs", "acs1"};
```

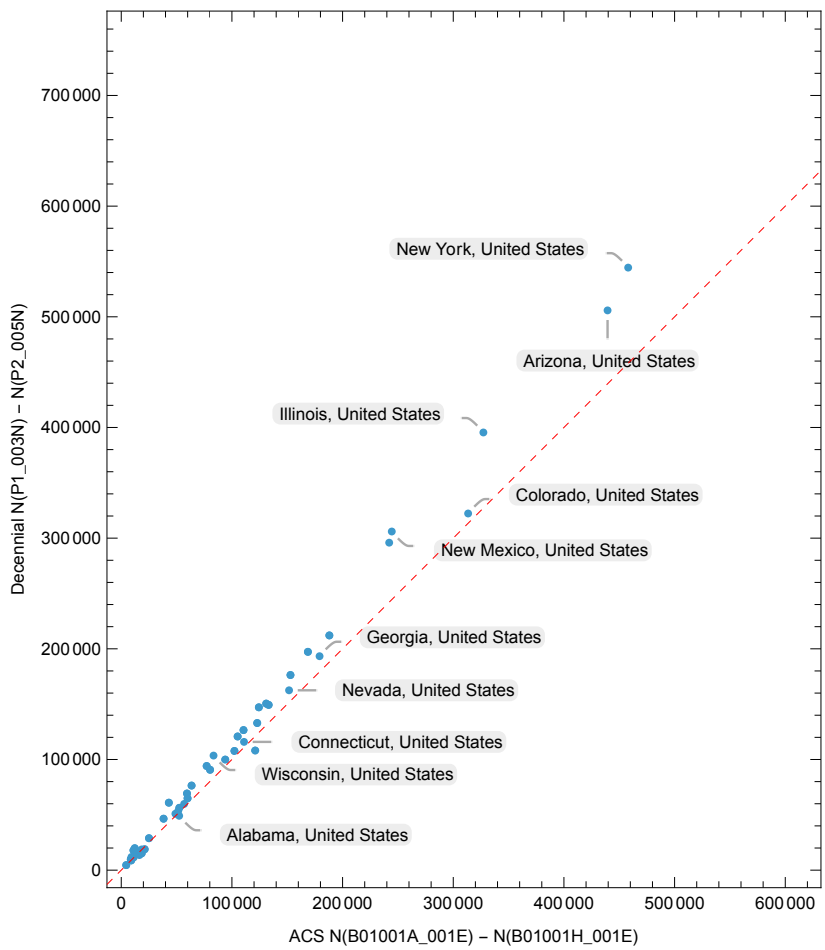
In[128]:=

```
stateCountsACS =
  allStateValues[{"B01001A_001E"}, pathACS] -
   allStateValues[{"B01001H_001E"}, pathACS];
stateCountsDec =
  allStateValues[{"P1_003N"}, pathDec] -
   allStateValues[{"P2_005N"}, pathDec];
```

In[130]:=

```
ListPlot[
 KeyMap[fipsStates, Merge[{stateCountsACS, stateCountsDec}, Identity]],
 Epilog → {Red, Dashed, InfiniteLine[{0, 0}, {1, 1}]},
 AspectRatio → Automatic,
 (*PlotRange→{{0,1300},{0,1300}},*)
 Frame → True,
 FrameLabel →
  {"ACS N(B01001A_001E) - N(B01001H_001E)", "Decennial N(P1_003N) - N(P2_005N)"}
]
```

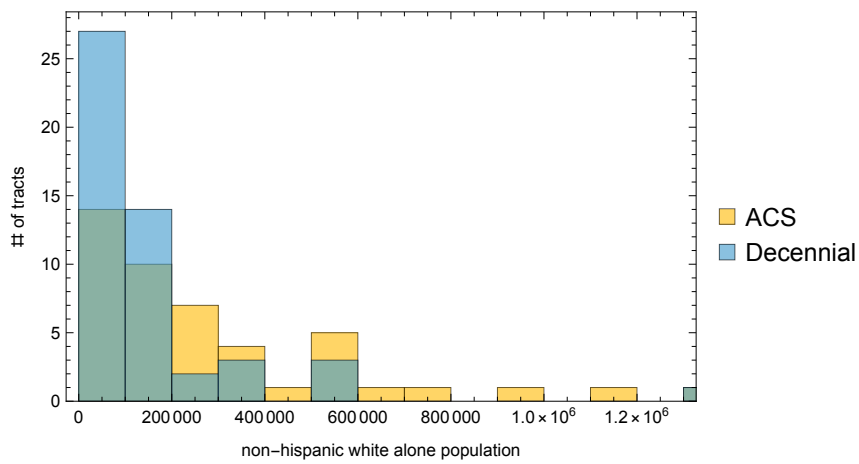
Out[130]=



In[145]:=

```
Histogram[{
  stateCountsACS,
  stateCountsDec
 },
 {100 000},
 ChartLegends → {"ACS", "Decennial"},
 Frame → True, FrameLabel → {"non-hispanic white alone population", "# of tracts"}
]
```

Out[145]=



2019 1-year

In[134]:=

```
pathACS = {"data", "2019", "acs", "acs1"};
```

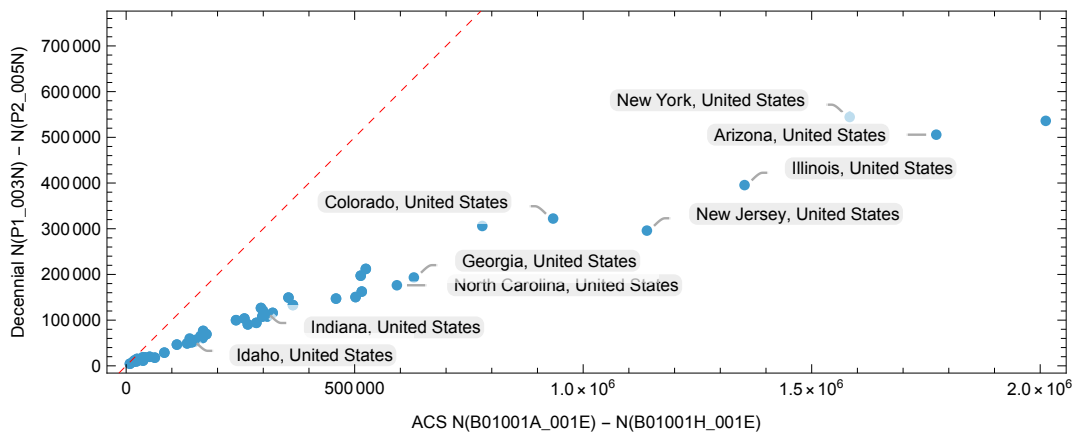
In[139]:=

```
stateCountsACS =
  allStateValues[{"B01001A_001E"}, pathACS] -
   allStateValues[{"B01001H_001E"}, pathACS];
stateCountsDec =
  allStateValues[{"P1_003N"}, pathDec] -
   allStateValues[{"P2_005N"}, pathDec];
```

In[141]:=

```
ListPlot[
 KeyMap[fipsStates, Merge[{stateCountsACS, stateCountsDec}, Identity]],
 Epilog → {Red, Dashed, InfiniteLine[{0, 0}, {1, 1}]},
 AspectRatio → Automatic,
 (*PlotRange→{{0,1300},{0,1300}},*)
 Frame → True,
 FrameLabel →
  {"ACS N(B01001A_001E) - N(B01001H_001E)", "Decennial N(P1_003N) - N(P2_005N)"}
]
```

Out[141]=



Old versions